

MERN Stack Learning Management System

Abstract

This project is a web-based Learning Management System (LMS) developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js) with Tailwind CSS for user interface styling. The system is designed to connect students, tutors, and administrators in a single platform where educational content can be shared, requested, approved, and accessed securely. Students can browse free and paid courses, request access to paid content, and manage their learning library. Tutors can upload courses and approve or reject student requests, while administrators oversee the entire system. This project demonstrates full-stack development, role-based access control, RESTful API design, and modern UI development.

Introduction

With the rapid growth of online education, there is a strong need for efficient platforms that allow secure content sharing and controlled access to paid educational resources. This project aims to solve this problem by providing a role-based Learning Management System that supports students, tutors, and administrators. The system ensures that paid content is only accessible after tutor approval while allowing free content to be accessed instantly.

Objectives

- To develop a full-stack LMS using the MERN stack.
- To implement role-based authentication and authorization.
- To allow tutors to manage educational content.
- To allow students to request and access courses securely.
- To provide an admin dashboard for system management.

Technology Used

Frontend:

- React.js
- Tailwind CSS

Backend:

- Node.js
- Express.js

Database:

- MongoDB

Other Tools:

- Axios

- JWT Authentication
- Git & GitHub

System Architecture

The system follows a client-server architecture. The frontend is developed using React.js and communicates with the backend via RESTful APIs. The backend, built with Node.js and Express.js, handles authentication, authorization, and business logic. MongoDB is used as the database to store user, content, and request data.

User Roles and Features

Admin:

- View and manage all users
- View all courses

Tutor:

- Add, update, and delete courses
- Approve or reject student access requests

Student:

- View free courses
- Request access to paid courses
- Access approved courses

Database Design

The database consists of multiple collections including Users, Contents, Requests, and Reviews. MongoDB schemas are used to define relationships between students, tutors, and content.

Instructions / How to Use the System

1. Start the backend server using Node.js.
2. Start the frontend using Vite and React.
3. Register or log in as a Student, Tutor, or Admin.
4. Tutors can upload content from their dashboard.
5. Students can browse courses and request access to paid content.
6. Tutors approve or reject requests.
7. Approved courses appear in the student's library.

Conclusion

This project successfully implements a role-based Learning Management System using modern web technologies. It demonstrates practical use of the MERN stack, secure authentication, and clean UI design using Tailwind CSS.

Future Enhancements

- Online payment integration
- Live classes and video streaming
- Certificate generation
- Mobile application support

References

- MongoDB Documentation
- React.js Documentation
- Node.js Documentation
- Express.js Documentation